Setting Up and Running KNN_CUDA with Singularity on Toubkal GPUs

1. Pull the NVIDIA Container

singularity pull docker://nvcr.io/nvidia/ai-workbench/python-cuda117:1.0.6

2. Create and Navigate to the Shared Directory

mkdir shared

cd shared

3. Clone the KNN_CUDA Repository

git clone https://github.com/unlimblue/KNN_CUDA

4. Configure Singularity Bind Paths

vim ~/.bashrc

export SINGULARITY_BINDPATH=(path of your shared directory):/mnt

source ~/.bashrc

5. Allocate GPU Resources

salloc -t 7:00:00 -n 1 -A sw_stack-373lcd9r8io-default-gpu -p gpu_100 --gres=gpu:1

6. Run the Singularity Container

echo "export PATH=/mnt/bin:\$PATH" >> ~/.bashrc && source ~/.bashrc

echo 'export TORCH_CUDA_ARCH_LIST="8.0"' >> ~/.bashrc && source ~/.bashrc

cd ..

singularity run --nv python-cuda117_1.0.6.sif

7. Set Up the KNN_CUDA Environment

cd /mnt/

8. Build and Install KNN_CUDA

cd KNN_CUDA/

chmod +x ninja

Edit the Makefile:

```
.PHONY : build reqs install clean
NINJA := $(shell command -v ninja 2> /dev/null)


build : reqs
	python3 setup.py bdist_wheel

reqs :
ifndef NINJA
	mkdir -p /mnt/bin
	cp ./ninja /mnt/bin/ninja
endif
	pip3 install -r requirements.txt

install :
	pip3 install --upgrade dist/*.whl

clean :
	-rm -rf build dist/* *.egg-info
```

9. Compilation and installation

make && make install

10. Create test directory and execute

mkdir tests

cd tests

vim test.py (import any knn cuda code to test it)

python3.10 test.py